

Gemma 4 E2B as an End-to-End Drone Navigation Controller: A Pilot Trial in the 25-VLM Lineup

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****Gemma 4 E2B as an End-to-End Drone Navigation Controller:****

****A Pilot Trial in the 25-VLM Lineup****

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*****Abstract*****

*We extend the 25-VLM closed-loop benchmark from ``Closing_the_Metric_Gap`` with one additional architecture: Gemma 4 E2B, the multimodal Gemma 4 variant released by Google DeepMind and demonstrated by NVIDIA on Jetson Orin Nano (5.1 B parameters / 2.3 B effective via Per-Layer Embeddings, 35 layers, 128 K context, multimodal text+image+audio). Gemma 4 was deployed as a direct image+text→goal controller on the same Isaac Sim benchmark used in our prior work (51 tasks × 3 environments, 1 seed = 153 trials, identical task distribution to Track A v6). Aggregate final error is ****47.78 m****, far worse than the 9.50 m Hover baseline and consistent with the paper's central finding that no end-to-end 7–8 B-class VLM reliably outperforms doing nothing. Step-1 (single-shot) prediction error is 10.66 m and improvement ratio is 0.65×, indicating significant divergence between single-prediction accuracy and closed-loop trajectory. Direction-cosine accuracy is +0.11 — substantially below the 0.83–0.91 typical of frontier 7–8 B VLMs, suggesting that at the 2.3 B-effective scale the directional half of the metric gap also opens up. Per-tier results match expectation: trivial tiers (``1_stationary``, ``2_cardinal`` step-1) work perfectly; visually grounded tiers (``6_visual`` 92.7 m, ``10_occluded`` 153.0 m) fail catastrophically; semantic tiers (``12_semantic`` 5.95 m) survive at the threshold level. End-to-end mean inference latency is 17.8 s/step, which makes Gemma 4 unsuitable as a real-time controller in this configuration without aggressive quantization or token caps. The same Gemma 4 weights, deployed inside the modular Track A pipeline as a target-identifier (replacing Nemotron+GDINO+Florence, evaluated 2026-04-05 prior to this report), score 9.35 m aggregate — beating hover by 0.15 m (n.s.) and beating end-to-end Gemma 4 by 38.4 m. The split is the cleanest single-model demonstration of the separation principle in our lineup to date.*

****I. CONTEXT****

``Closing_the_Metric_Gap`` evaluated 25 VLM architectures across 10,200 closed-loop quadrotor trials in NVIDIA Isaac Sim and found that no end-to-end 7–8 B VLM reliably outperforms hovering in place. The diagnosis was a ****metric gap**** — VLMs achieve 0.83–0.91 directional cosine accuracy but 6–10 m distance error on 4–12 m targets — and the prescription was the ****separation principle****: use VLMs only for semantic target identification and delegate spatial grounding to dedicated depth and detection modules.

``Engineering_the_Separation_Principle`` then iterated the modular Track A pipeline through 18 refinements (R3–R18) and made the lineage of design decisions explicit. A natural follow-up question, raised by the publication of Google DeepMind's Gemma 4 [1] and NVIDIA's Jetson Orin Nano deployment write-up [2], is: ***does the metric gap diagnosis still hold for a brand-new 5 B-parameter (2.3 B-effective) instruction-tuned multimodal model designed explicitly for on-device perception?*** This pilot answers that question for the end-to-end deployment.

****II. METHOD****

Model. Gemma 4 E2B-it (``google/gemma-4-E2B-it``, total 5.1 B / effective 2.3 B via Per-Layer Embeddings, 35 layers, 128 K context, native multimodal head). Loaded from the canonical HuggingFace repo at bf16 (10.25 GB). VRAM at inference time was ~25 GB (transformers 5.5.4 / torch 2.11.0 / cuda:0); this is significantly above the 9.5 GB observed in the Track A target-selector configuration of 2026-04-05 and reflects the larger generation buffer required by ``max_new_tokens=300``. The Q4_K_M GGUF quantization recommended by NVIDIA's Jetson deployment was not used in this run; cross-quantization comparison is left to a follow-up.

Deployment. Gemma 4 requires transformers 5.x, which conflicts with Isaac Sim's bundled torch / typing-extensions stack used by the benchmark runner. We therefore run the model in a separate venv (~/gemma4-venv) inside `gemma4\_server.py` and talk to it from a thin `VLMGemma4` architecture class (benchmark/architectures/vlm_gemma4.py) over a Unix-domain socket. The same socket already serves the Track A modular pipeline; a `mode: "navigate"` field selects the end-to-end path. All other components of the closed-loop run — Isaac Sim 5.1, Pegasus 5.1.0, the Pegasus quadrotor, the 60 s budget, the 2 s flight segments, and the 51-task / 3-environment task distribution — are exactly those of the Track A v6 5-seed run on 2026-04-05.

Prompt and parsing. The prompt mirrors `vlm\_smolvlm.py` for direct comparability with the existing lineup: drone-frame coordinates (+x forward, +y left, +z up), free-form chain-of-thought, and a JSON-only output contract `{x, y, z, reasoning}`. Single-pass greedy decoding (do_sample=False, max_new_tokens=300). Failure-mode parsing falls back to extracting the first three numbers if JSON parsing fails. No few-shot examples, no detection prepass, no depth conditioning — this is the unmediated end-to-end test.

Scope. 1 seed × 51 tasks = 153 trials (matches one seed of the Track A v6 reference run). The 25-VLM lineup of `Closing\_the\_Metric\_Gap` used 8 seeds; the 1-seed numbers below are therefore preliminary point estimates suitable for ranking-level claims, not for tight CIs. Wall time for the full Isaac Sim run was 48.6 minutes.

III. RESULTS

Aggregate. Final position error 47.78 m, step-1 prediction error 10.66 m, improvement ratio 0.65× (the closed-loop trajectory is *worse* than the single-shot prediction), direction cosine +0.11, success@5 m = 18%, success@10 m = 35%, mean collisions 14.0, collision-free rate 33%, mean inference 17.8 s/step. ****Hover baseline** (same task distribution): 9.50 m, +0.00 dir, success@5 m = 16%, 100% CF. ****Random baseline**: 39.06 m. Track A v6 (Gemma 4 as target-selector inside the modular pipeline, 2026-04-05): 9.35 m, +0.07 dir, 71% CF.

Per-tier breakdown (n trials, final / step-1 / s1→final ratio / direction):

Tier	n	Final	Step-1	s1/final	Dir
1_stationary	3	0.18m	0.18m	1.00×	+1.00
2_cardinal	3	34.15m	2.56m	0.08×	+0.67
3_named_near	3	5.04m	5.03m	1.00×	+0.23
4_named_far	3	45.84m	13.62m	0.30×	+0.34
5_spatial	3	21.48m	11.61m	0.54×	−0.30
6_visual	3	92.67m	6.72m	0.07×	+0.58
7_reasoning	6	18.38m	17.62m	0.96×	−0.27
8_multi_leg	3	56.54m	10.18m	0.18×	−0.49
9_obstacle	6	49.44m	10.61m	0.21×	+0.15
10_occluded	6	152.96m	23.13m	0.15×	−0.10
11_appearance	3	43.86m	3.12m	0.07×	+0.64
12_semantic	3	5.95m	5.49m	0.92×	−0.18
13_counting	3	22.89m	10.80m	0.47×	+0.02
14_exploration	3	42.07m	9.17m	0.22×	−0.18

IV. ANALYSIS

The metric gap is present and the directional half is also wide. On the tiers where step-1 prediction is reasonable (e.g. `2\_cardinal` 2.56 m, `11\_appearance` 3.12 m, `6\_visual` 6.72 m), the closed-loop trajectory diverges by up to 14×: step-1 → final ratios of 0.07–0.18× for cardinal, appearance, and visual tiers. This is the same replanning-divergence phenomenon characterized in `Closing\_the\_Metric\_Gap` Sec. IV: a single-shot estimate is rolled into a sequence of small flight segments, each re-querying the VLM, and the cumulative coordinate drift dominates the budget. What is new here is that direction cosine has dropped from the 0.83–0.91 plateau seen at 7–8 B to +0.11 aggregate, with several tiers (`5\_spatial`, `7\_reasoning`, `8\_multi\_leg`, `12\_semantic`) actually anti-correlated. The metric gap is no longer the only failure mode at this scale — directional reasoning has also regressed.

Trivial tiers and semantic tiers are the survivors. `1\_stationary` is solved by every model that returns (0,0,1.5) on "Hover in place," and Gemma 4 does so. `12\_semantic` (5.95 m, 100% success@10 m) is the only

non-trivial tier where Gemma 4 holds up; the 2026-04-05 Track A target-selector pass also performed well on semantic tasks, consistent with the separation principle's prediction that VLM-as-semantic-classifier is the regime where these models actually contribute.

The same model, deployed two different ways, lands 38.4 m apart. Gemma 4 inside Track A: 9.35 m. Gemma 4 end-to-end: 47.78 m. The 4× gap is not an artifact of seeds (Track A v6's CI is [8.53, 10.24]; end-to-end's score sits four CIs above) and is not explained by any difference in task distribution, simulator state, prompt, or weights — those are held constant. The only difference is whether the VLM is asked to output a goal coordinate or a target name. This is the cleanest single-model demonstration of the separation principle in our lineup to date.

Latency. 17.8 s/step end-to-end is far above the 0.92 s observed on the smoke test and the 1.4 s observed on synthetic prompts. The gap reflects long generation traces on the harder tiers (the model often emits 200+ tokens of reasoning before the JSON, hitting the 300-token cap on `4\_named\_far`, `7\_reasoning`, and `10\_occluded`). At this latency, the Pegasus 60 s budget allows only ~3 closed-loop queries per trial, which compounds the divergence problem.

V. WHERE THIS SITS IN THE PAPER LINEUP

If folded into the `Closing\_the\_Metric\_Gap` Table I (the 25-VLM aggregate), Gemma 4 E2B end-to-end would be the smallest serious VLM in the row below SmolVLM 2B and would not change the paper's central claim — no entry beats hover. If folded into `Engineering\_the\_Separation\_Principle` as a Track A v6 ablation, the 38.4 m end-to-end / modular gap is a clean illustration of the separation principle's lever arm and would slot naturally into Sec. III. We do not recommend folding the present 1-seed result into either paper without a multi-seed re-run; we do recommend recording the end-to-end / modular split as a standalone observation.

VI. ARTIFACTS

- Server: `/home/yusuf/zeroclaw-train/gemma4\_server.py` (extended 2026-04-28 with `mode: "navigate"`).
- Architecture: `~/code/ishmael/benchmark/architectures/vlm\_gemma4.py`.
- Registration: `vlm\_gemma4` entry in `~/code/ishmael/benchmark/config/architectures.yaml`.
- Weights: `~/models/gemma-4-E2B-it/` on hoopoe (10.25 GB, bf16).
- Results JSON: `~/code/ishmael/results/20260428\_232405.json`.
- Runtime log: `~/gemma4\_full\_1seed.log`.
- Track A v6 reference: `~/code/ishmael/results/track\_a\_v6\_benchmark\_20260405\_162102.json`.

REFERENCES

- [1] Google DeepMind, "Gemma 4 model family," 2026, <https://huggingface.co/google/gemma-4-E2B-it>.
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- [3] Y. Saib, "Closing the Metric Gap: From Diagnosis to Solution in Vision-Language Drone Navigation," Astral Technology Corporation, 2026.
- [4] Y. Saib, "Engineering the Separation Principle: From Modular Architecture to Deployable Drone Navigation," Astral Technology Corporation, 2026.